EXAM PAPERS PRACTICE

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Detailed mark scheme

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AQA A Level Further Mathematics (7367)

Topic **D: Further Algebra and Functions**

Sub topic **D9: Rational Inequalities**

Mark Scheme

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Topic	**D: Further Algebra and Functions**
Sub topic	**D9: Rational Inequalities**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

(a) Asymptotes

$x = -1$

$x = -1$ OE

B1

$x = 2$

$x = 2$ OE

B1

$y = 0$

$y = 0$

B1

3

(b) $-\frac{1}{2} = \frac{x}{x^2 - x - 2} \Rightarrow x^2 - x - 2 = -2x$

Correctly removing brackets and fractions to reach $x^2 - x - 2 = -2x$ OE

M1

$x^2 + x - 2 = 0 \Rightarrow x = 1, x = -2$

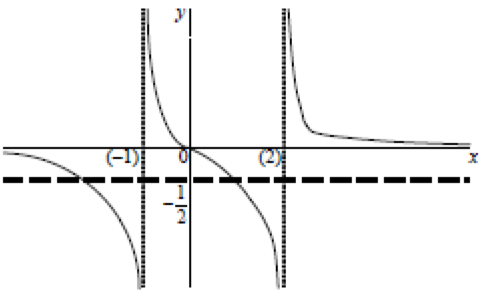
Correct two values for x-coordinates.

NMS 2 or 0 marks

A1

2

(c)



Three branches shown on sketch of C with either middle branch or outer two branches correct in shape

M1

All three branches, correct shape and positions and approaching correct asymptotes in a correct manner.

*If middle branch does **clearly** not go through the origin,* *then A0*

A1

Correct sketch of line (L), y = − 0.5 identified

B1

3

(d) $-2 \le x < -1$

Condone

B1

$1 \le x < 2$

Condone

B1

$-2 \le x < -1,\ 1 \le x < 2$

All complete and correct

B1

3

[11]

Q2.

(a) (i) Asymptotes $x = -2$, $x = 2$, $y = 0$

B1 × 3

3

(ii) Middle branch generally correct

Allow if max pt not in right place

B1

Other branches generally correct

B1

All branches approaching asymps

B1

Intersection at $\left(0, -\frac{1}{4}\right)$ indicated

Asymps must be shown correctly on diagram or elsewhere; B0 if any other intersections are shown

B1

4

(b) $y = -2$ when $x = \pm\sqrt{3.5}$

Allow NMS

B1

Sol'n $-2 < x < -\sqrt{3.5}, \sqrt{3.5} < x < 2$

Condone dec approx'n for $\sqrt{3.5}$ *;*
B1 if ≤ used instead of <

B2,1

3

[10]

Q3.

(a) (i) Asymptotes $x = 3$ and $y = 0$

may appear on graph

B1,B1

2

(ii) Complete graph with correct shape

B1

Coordinates $\left(0, -\frac{1}{3}\right)$ shown

B1

2

(iii) Correct line, (0, –5) and (2.5, 0) shown

B1

1

(b) (i) $2x^2 - 11x + 14 = 0$

B1

$x = 2$ or $x = 3.5$

M1 for valid method for quadratic

M1A1

3

(ii) $2 < x < 3$, $x > 3.5$

B1 for partially correct solution;
ft incorrect roots of quadratic
(one above 3, one below 3)

B2,1F

2

[10]

Q4.

(a) (i) Asymptotes $x = 0$, $x = 2$, $y = 1$

B1 × 3 **3**

(ii) Intersections at (1, 0) and (3, 0)

B1 **1**

(iii) At least one branch approaching asymptotes

B1

Each branch

B1 × 3 **4**

(b) $0 < x < 1$, $2 < x < 3$

Allow B1 if one repeated error occurs,
eg ≤ for <

B1,B1 **2**

Alternative:
Complete correct algebraic method

M1A1 **(2)**

[10]

Q5.

(a) Asymptotes $y = 0$, $x = –1$, $x = 1$

B1 × 3 **3**

(b) Three branches approaching two vertical asymptotes

Asymptotes not necessarily drawn

B1

Middle branch passing through O

with no stationary points

B1

Curve approaching $y = 0$ as $x \to \pm\infty$

B1

All correct

with asymptotes shown and curve

approaching all asymptotes correctly

B1

4

(c) Critical values x = –1, 0 and 1

B1

Solution set $-1 < x < 0$, $x > 1$

M1 if one part correct
or consistent with c's graph

M1A1

3

[10]

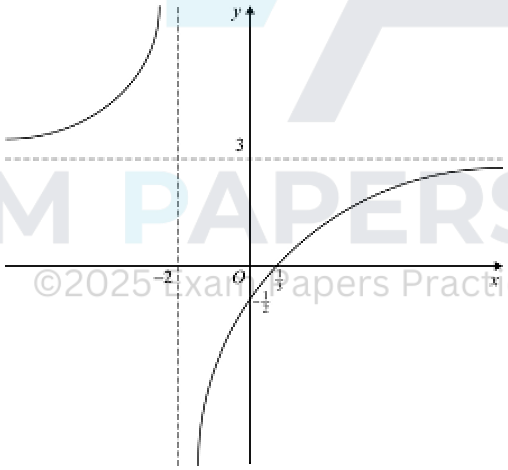
Q6.

(a) Asymptotes $x = -2$, $y = 3$

B1,B1

2

(b)



Curve approaching asymptotes

B1

Passing through $\left(\frac{1}{3}, 0\right)$ *and* $\left(0, -\frac{1}{2}\right)$

B1,B1

Both branches generally correct
B1 if two branches shown

B1,B1

5

(c) Solution set is $x > \frac{1}{3}$

B1 for good attempt;

ft wrong point of intersection

B2,1F

2

[9]

Q7.

(a) Asymptotes $x = 1$, $y = 6$

B1B1

2

(b) Curve (correct general shape)

M1

Curve passing through origin

A1

Both branches approaching $x = 1$

A1

Both branches approaching $y = 6$

A1

SC Only one branch:
B1 for origin
B1 for approaching both asymptotes
(Max 2/4)

4

(c) Correct method

M1

Critical values ±1
From graph or calculation

B1B1

Solution set $-1 < x < 1$
ft one error in CVs; NMS
4/4 after a good graph

A1ft

4

[10]

Examiner reports

Q1.

In part (a), the equations of the vertical asymptotes were usually stated correctly, but the equation $y = 0$ of the horizontal asymptote was occasionally omitted or an incorrect equation, usually $y = 1$, was given. The very few students who failed to express the asymptote in the form of an **equation** gained no credit. In part (b), students generally understood the requirements but algebraic errors in rearrangement and expansion of brackets sometimes led to incorrect quadratic equations and solutions. In part (c), the curve was generally well sketched showing the main features of the three branches. The asymptotes were usually included which significantly reduced the number of times a mark was lost due to overlapping branches. The examiners also expected to see the central branch passing through the origin which was not always the case in students' sketches. The behaviour of the curve as it approached asymptotes was generally appropriate. The line $y = –1 / 2$ was not always clearly defined either by scale or label and surprisingly even oblique lines were occasionally seen.

In part (d), average and better students often gained two marks but some lost the third mark because they did not use strict inequality signs at the –1 and 2 boundaries. A large proportion of students found this final part challenging and some did not use the graph they had drawn in part (c), preferring to embark on a fresh, complex, algebraic approach to solving the inequality which rarely led to success.

Q2.

The asymptotes asked for in part (a)(i) were usually given correctly, and there was a high standard of curve sketching in part (a)(ii). In part (b), only a small proportion of candidates gave the full correct solution of the inequality. As well as looking at the sketch-graph, it was necessary to do a calculation here, and many candidates earned one mark out of three by finding the two correct points of intersection of the curve with the line $y = –2$.

Q3.

The first eight marks out of the ten available in this question were gained without much difficulty by the majority of candidates, apart from some careless errors such as omitting to indicate the coordinates asked for on the sketch.

By contrast the inequality in part (b)(ii) was badly answered. Few candidates seemed to think of reading off the answers from the graph, the majority preferring an algebraic approach, which if done properly would have been worth much more than the two marks on offer. The algebraic method usually failed at the first step with an illegitimate multiplication of both sides of the inequality by $x – 3$. Some candidates multiplied by $(x – 3)^2$ but could not cope with the resulting cubic expression.

Q4.

The first two parts of this question were generally well answered, and there were many good sketches in part (a)(iii), though the middle branch of the curve was often badly drawn, many candidates appearing to ignore the helpful information provided about there being no stationary points. Part (b) could be answered very easily by looking at the parts of the graph which were below the x-axis, and reading off the two intervals in which this was the case. Instead, many candidates used a purely algebraic approach, and in most cases the inequality was rapidly reduced to the incorrect form $(x – 1)(x – 3) < 0$, though some candidates obtained the correct version, $x(x – 1)(x – 2)(x – 3) < 0$, and after making a table of signs came up with the correct solution.

Q5.

Most candidates scored well here, picking up marks in all three parts of the question. Those who failed to obtain full marks in part (a) were usually candidates who gave $y = 1$ instead of $y = 0$ as the equation of the horizontal asymptote. The sketch was usually reasonable but some candidates showed a stationary point, usually at or near the origin, despite the helpful information given in the question. There were many correct attempts at solving the inequality in part (c), though some answers bore no relation to the candidate's graph.

Q6.

Many candidates scored well on parts (a) and (b) of this question, but relatively few candidates made a good attempt at part (c). In part (a) the asymptotes were usually found correctly, as were the coordinates of the required points of intersection in part (b). The graph in part (b) was usually recognisable as a hyperbola, or at least as one branch of a hyperbola, the other branch not being seen. In part (c) many candidates resorted to algebraic methods for solving inequalities rather than simply reading off the solution set from the graph. These algebraic methods, more often than not, were spurious.

Q7.

Parts (a) and (b) of this question were generally well answered, though some otherwise good graphs did not pass through the origin as required. The inequality in part (c) was not tackled at all well. Few candidates seemed to realise that a good graph allowed them to read off the answer, the only further detail required being to ascertain the value of x for which $y = 3$. A distressingly large number of candidates simply multiplied both sides of the inequality by the denominator $(x - 1)$. Those who adopted sounder methods often arrived at the statement $x^2 < 1$, but then showed a lack of mathematical awareness by drawing a meaningless conclusion such as $x < \pm 1$.